MA 2103(3)/MA 2113(3) Mathematical Statistics

Class Test

Date: 09-09-26 Time: 45 min Marks: 30

Answer all the questions.

1. (2 points) State Chebyshev's inequality explaining all the notations used in the statement.

2. The table below shows the joint distribution of two random variables, X and Y.

		Values of Y			
		1	2	3	4
Values of X	1	$6c$	$3c$	$2c$	$4c$
	2	$4c$	$2c$	$4c$	0
	3	$2c$	c	0	$2c$

   (a) (3 points) Prove that $c = \frac{1}{30}$ and calculate the marginal distributions of X and Y.

   (b) (6 points) Calculate $E(X)$ and $\mathrm{Var}(X)$ and show that $\mathrm{Cov}(X, Y) = 0$.

   (c) (1 point) State, with a reason, whether or not X and Y are independent.

3. (7 points) The probability density function of X is given by

$$f(x) = \begin{cases} \frac{2}{3}, & 0 \le x \le 1 \\ \frac{1}{3}, & 1 \le x \le 2 \\ 0, & \text{elsewhere.} \end{cases}$$

   Find the moment generating function of X. Hence obtain the mean and variance of X.

4. The joint probability density function (pdf) of random variables X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{2} x e^{-y} & 0 < x < 2, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

(a) (5 points) Obtain the marginal pdfs of X and Y. Are X and Y independent?

(b) (4 points) Find the conditional pdf $f(y \mid x)$. Obtain $P\big(0 < Y < \frac{1}{2} \mid X = 1\big)$.

(c) (2 points) Find $E(Y \mid X = 1)$.